CANONICAL 2-ARROW MAKING THE QUOTIENT-STACK SQUARE 2-COMMUTATIVE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 044R: sources/stacks-chow/tags.txt, source lines 5410–5410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Assumptions and notation as in Lemma 044R. There exists a canonical* 2*-morphism* $\alpha : \pi \circ s \to \pi \circ t$ *making the diagram*

$$\begin{array}{ccc} \mathcal{S}_R & \xrightarrow[s]{} & \mathcal{S}_U \\ {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle \pi} \\ \mathcal{S}_U & \xrightarrow{\pi} & [U/R] \end{array}$$

2*-commutative.*

Proof. Let S' be a scheme over S. Let $r : S' \to R$ be a morphism over S. Then $r \in R(S')$ is an isomorphism between the objects $s \circ r, t \circ r \in U(S')$. Moreover, this construction is compatible with pullbacks. This gives a canonical 2-morphism $\alpha_p : \pi_p \circ s \to \pi_p \circ t$ where $\pi_p : \mathcal{S}_U \to [U/_pR]$ is as in the proof of Lemma 044R. Thus even the diagram

$$\begin{array}{ccc} \mathcal{S}_R & \xrightarrow[s]{} & \mathcal{S}_U \\ \downarrow{\scriptstyle t} & & {\scriptstyle \pi_p}\downarrow \\ \mathcal{S}_U & \xrightarrow{\pi_p} & [U/_pR] \end{array}$$

is 2-commutative. Thus a fortiori the diagram of the lemma is 2-commutative. □

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